

2. Problem 3 - Computing a Recurrence

Suppose a dynamic-programming algorithm takes as input a number x and returns $f(x - 1)$, where

$$f(x) = \begin{cases} i \cdot f(i - 1) + f(i - 2), & \text{if } i \geq 2 \\ 1, & \text{otherwise} \end{cases}$$

- (a) Create a table showing the values computed when $x = 5$.
- (b) Write pseudocode for a dynamic-programming algorithm that stores all intermediate values and works for any positive x .

Solution: Table for $x=5$ (need $f(4)$):

i	$f(i)$
0	1
1	1
2	$2 \cdot 1 + 1 = 3$
3	$3 \cdot 3 + 1 = 10$
4	$4 \cdot 10 + 3 = 43$

$$f(4) = 43$$

Pseudocode:

```
function computeRecurrence(x):  
    f[0] = 1  
    f[1] = 1  
    for i from 2 to x-1:  
        f[i] = i * f[i-1] + f[i-2]  
    return f[x-1]
```